Graduate students are given a solid background in the fundamentals of classical and modern physics at an advanced level and are encouraged early in their studies to learn about the research interests of the faculty so that a thesis topic can be identified.

Primary Contact

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Physics Department
https://physics.mines.edu/
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Program Requirements

Students entering graduate programs in the Physics Department will select an initial program in consultation with the departmental graduate student advising committee until such time as a research field has been chosen and a thesis committee appointed.

Master of Science

Requirements:

MS-Thesis in Applied Physics: 18 credits of course work in an approved program, 2 credits of seminar, plus 10 credits of research credit, with a satisfactory thesis.

MS-non-Thesis in Applied Physics: 28 credits of course work and 2 credits of seminar. At least 21 of the required 30 credits of course work must be taken as a registered master's degree student at Mines. The program will allow up to 7 credits of independent study coursework to count towards the non-thesis MS degree. Up to 9 credits of graduate courses may be transferred into the degree program, provided that those courses have not been used as credit toward a Bachelor's degree. The student's committee makes decisions on courses to be taken, transfer credit, and examines the student's written report and oral presentation resulting from an independent study.

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Doctor of Philosophy

Requirements: 32 credits of coursework in an approved program plus 40 credits of research credit, with a satisfactory thesis. 12 credits of course work will be in a specialty topic area defined in consultation with the thesis advisor. Possible specialty topic areas within the Physics Department exist in Optical Science and Engineering, Condensed Matter Physics, Theoretical Physics, Renewable Energy Physics, and Nuclear/Particle Physics and Astrophysics.

To demonstrate adequate preparation for the PhD degree in Physics, each student must achieve a grade point average of 3.0 or better in the core courses. Students not meeting this standard must pass oral examinations covering the relevant course content or retake the courses within one year. This process is part of the requirement for admission to candidacy, which full time PhD students must complete within two calendar years of admission, as described in the campus-wide graduate degree requirements (p. 29) section of this bulletin. Other degree requirements, time limits, and procedural details can be found in the Physics Department Graduate Student Advising Brochure.

Physics Colloquium

All full-time physics graduate students must attend the Physics Colloquium, which is represented in the curriculum by the Graduate Seminar courses. Students must take one of these courses every semester that they are enrolled at Mines. Those students who are in the MS Program, sign up for PHGN501 (fall) and PHGN502 (spring). Students in the PhD program sign up for PHGN601 (fall) and PHGN602 (spring). At the end of each semester students are assigned either a satisfactory or unsatisfactory progress grade, based on attendance, until the final semester of the student's degree program, when a letter grade is assigned based on all prior semesters' attendance grades. As a result, while these courses are taken each year, only 1 hour total of course credit is conferred for each of 501, 502, 601, or 602. Students who have official part-time status and who have already taken at least one semester of 501 and 502 for the MS degree, or 601 and 602 for the PhD degree are not required to sign up for Graduate Seminar during subsequent semesters.

Prerequisites

The Graduate School of Colorado School of Mines is open to graduates from four-year programs at accredited colleges or universities. Admission to the Physics Department MS and PhD programs is competitive and is based on an evaluation of undergraduate performance, standardized test scores, and references. The undergraduate course of study of each applicant is evaluated according to the requirements of the Physics Department.

Required Curriculum

Master of Science (with Thesis) in Applied Physics

Core Courses		
PHGN511	MATHEMATICAL PHYSICS	3.0
PHGN515	GRADUATE LABORATORY	3.0
PHGN520	QUANTUM MECHANICS I	3.0
PH ELECT	Electives **	9.0
PHGN501 & PHGN502	GRADUATE SEMINAR and GRADUATE SEMINAR *	2.0
PHGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	10.0
Total Semester Hrs		**30.0**

* Graduate Seminar: Each full-time MS graduate student will register for Graduate Seminar each semester for a total of 2 cumulative credits over the degree.

** Electives maybe chosen from classes taught in physics or relevant classes taught in other departments, as approved by the graduate advisor and thesis committee.

Master of Science (non-Thesis) in Applied Physics

Core Courses		
PHGN511	MATHEMATICAL PHYSICS	3.0

PHGN515	GRADUATE LABORATORY	3.0
PH ELECT	Electives **	22.0
PHGN501 & PHGN502	GRADUATE SEMINAR and GRADUATE SEMINAR *	2.0
Total Semester Hrs		**30.0**

* Graduate Seminar: Each full-time MS graduate student will register for Graduate Seminar each semester for a total of 2 cumulative credits over the degree.

** Electives may be chosen from additional classes in physics, other disciplines relevant to their focus, and one of CSM 501 (Grad Student Skills), SYGN 501 (research skills for graduate students), or responsible conduct of research (PHGN 503 or SYGN 502).
At least one elective must be in a course with a significant laboratory or design emphasis. Courses with a laboratory component include, PHGN 532 Low Temp Microwave measurement, PHGN 535 Microprocessing lab, and PHGN 581 laser physics. Coursework with heavily applied components include, MLGN502 Solid State Physics, PHGN 504 Radiation Detection, PHGN 519 Fundamentals of Quantum Information, PHGN 542 Solid State Devices, PHGN 566 Modern optical Engineering. Other classes that fulfil this requirement may be approved by the graduate advisor.
The program will allow up to 7 credits of independent study coursework to count towards the non-thesis MS degree.

Doctor of Philosophy in Physics

Core Courses		
PHGN507	ELECTROMAGNETIC THEORY I	3.0
PHGN511	MATHEMATICAL PHYSICS	3.0
PHGN515	GRADUATE LABORATORY	3.0
PHGN520	QUANTUM MECHANICS I	3.0
PHGN521	QUANTUM MECHANICS II	3.0
PHGN530	STATISTICAL MECHANICS	3.0
PHGN601 & PHGN602	ADVANCED GRADUATE SEMINAR and ADVANCED GRADUATE SEMINAR *	2.0
PH ELECT	Special topic area electives	12.0
PHGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	40.0
Total Semester Hrs		**72.0**

* Graduate Seminar: Each full-time PhD graduate student will register for Graduate Seminar each semester for a total of 2 cumulative credits over the degree.

Fields of Research

Applied Optics: lasers, ultrafast optics, non-linear optics, laser-produced plasmas, micromachining, multiphoton microscopy, structured illumination microscopy, metasurface design and fabrication, high harmonic generation, multidimensional spectroscopy, novel phase resolved imaging techniques.

Subatomic: low energy nuclear structure and astrophysics, applied nuclear physics, high-energy cosmic-ray and neutrino physics, neutrinoless double beta decay, low-threshold detector development.

Condensed Matter and Materials Physics: photovoltaics, thermoelectrics, plasmonics, materials discovery, thin-film semiconductors, amorphous materials, excitonic materials, optoelectronic materials, nanostructures and quantum dots, self-assembled systems, organic and soft condensed matter, x-ray diffraction, Raman spectroscopy, x-ray photoelectron spectroscopy, Auger spectroscopy, scanning probe microscopies, atom probe tomography, first principles condensed materials theory**,** electronic structure, topological disorder.

Quantum Physics: quantum Information (quantum computing, quantum simulation, quantum error correction, superconducting and defect qubits), quantum many-body theory, disorder in quantum materials, quantum complexity, superconductivity, low-temperature physics, cryogenic electronics, strongly correlated systems.

GRADUATE CERTIFICATE IN OPTICS FOR ENGINEERING

Program Requirements

The graduate certificate program in Optics for Engineering is targeted to train recent graduates or midcareer professionals with a Bachelor of Science degree in physics, chemistry, materials science, electrical or mechanical engineering, or other related fields. The program will provide them with a basic knowledge of optics and lasers so they can apply it to the challenges and demands of advanced technologies that use optical systems. the program offers students a number of electives to choose from and to tailor their education to their interest.

Graduate Certificate Curriculum Requirements:

The certificate option consists of one core class, plus two additional electives, for a total of 9 credit hours. PHGN461 is considered a prerequisite for this certificate but is offered online in the summer, allowing students to easily fulfill this requirement. Students who have taken the equivalent of PHGN461 at a different institution can request to have this prerequisite waived.

Graduate Certificate

PHGN581	LASER PHYSICS	3.0
Electives	See Elective Listing	6.0

Coursework Details:

Students will need PHGN461 (or its equivalent) as a prerequisite for the certificate. This course is offered in the summer (online). PHGN581 is offered in the fall (in-person). Students will select two electives from the approved list below.

Approved Electives (select two):

PHGN570	FOURIER AND PHYSICAL OPTICS	3.0
PHGN566	MODERN OPTICAL ENGINEERING	3.0
PHGN585	NONLINEAR OPTICS	3.0
EENG528	COMPUTATIONAL ELECTROMAGNETICS	
GPGN570	APPLICATIONS OF SATELLITE REMOTE SENSING	3.0
EENG507	INTRODUCTION TO COMPUTER VISION	3.0
EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0